Live Fluorescence Imaging of Ion Transport and Synaptic Signaling in Transgenic Drosophila Larvae

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ABSTRACT
In this study, we explore the application of fluorescence microscopy for imaging neural patterns and systems within Drosophila larvae model organisms. Transgenic lines of drosophila expressed the presence of a highly specific protein fused to a variant of green fluorescent protein (GFP or GCaMP). Utilizing the Gal4/UAS system to generate fusion proteins of interest such as proteins prominent in sensory neurons and the Na^+/K^+ proton pump fused to standard GFP, Vglut-GCaMP, and DILP2-GCaMP (Deitcher, 2023). Fluorescent light emitted from the sample was captured with a CMOS camera and Pylon software (Johnson, 2025). Utilizing standard software such as ImageJ, the raw fluorescence of each image was quantified. The results suggest the strength of fluorescent microscopy for live imaging of long-term behavior in cells, such as sustained excitation or neurotransmitter release. GFP variant GCaMP was utilized to quantify neurotransmitter release as a function the Ca^{2+} concentration in the pre-synaptic site. These results specifically highlight the emphasis on extracellular K^+ concentration as a driving factor of neurotransmitter release and Ca^{2+} concentration in the presynaptic site. This paper then reflects on the strengths and limitations of 1-Photon Fluorescent Microscopy and explores the potential of 2-Photon Microscopy for future experiments.

INTRODUCTION
This paper utilized fluorescent microscopy as a means of exploring the active dynamics of various proteins critical for neural excitability and maintenance. Standard electrode-based techniques do not provide the specificity of proteins and neurons to record from. Fluorescence Microscopy allows for imaging of targeted proteins, found in highly specified neural pathways. By tagging specific proteins with GFP, only those proteins fluoresce when exposed to blue LED light (Wei et al., 2014). Upon illumination, a sample may either exhibit fluorescence due to electronic excitation and subsequent photon emission, or the sample may remain dark if no such excitation occurs. To enable targeted imaging, transgenic Drosophila larvae were genetically engineered to express GFP-fused proteins in select neurons and regions of interest. Without GFP, the native proteins and neurons of Drosophila larvae exhibit little to no fluorescence, enabling this specific imaging of GFP-bound proteins. Dr. David Deitcher provided the populations of five distinct fly lines which had been selectively determined to have GFP-variants fused to proteins of great interest in the neural network. In this study, we explore the imaging of these five transgenic drosophila larvae preparations. The targeted proteins are involved in sensory neurons, the sodium-potassium pump, and neurotransmitter release (both glutamate and octopamine). Vesicle-associated proteins proved to be extremely prominent in excitability when fused to GCaMP genes. We observed that altering the larvae environment through changes to the extracellular solution, we were able to drive changes in excitability and fluorescence. The fluorescence of these transport proteins proved dependent on both the receptor (protein-GCaMP) and ligand (Ca^{2+}) concentration in the presynaptic site. This approach combined both molecular specific and imaging, allowing for analysis into the underlying protein mechanism of excitation and neurotransmitter release in the system.

MATERIALS AND METHODS
A fluorescence microscope was utilized for imaging of each sample. Various proteins within specific samples were tagged with green fluorescent protein (GFP) through the introduction of a small DNA plasmids. These DNA sequences insert themselves into the host organism's genome at a specific recognized recombination site (Snapp, 2009). The model system was the *drosophila fly larvae*. The entire 15,000 gene genome has been sequenced, proving approximately 50% of fly proteins have human homologs (Deitcher, 2023). The ease of fly regeneration and development of transgenic families proves an efficient and accurate approach to altering and imaging the mechanisms of proteins in neural pathways. Drosophila fly larvae provide an insightful model system for much of vertebrate neurobiology.

Much of the methods for this experiment were derived from Johnson, 2025: *Imaging Guide for BION 4910.*

Transgenic Drosophila
We observe that various preparations of fly larvae were genetically modified to exhibit various proteins fused to green fluorescent proteins (GFPs). To produce these transgenic fly larvae, the GFP gene of interest was inserted downstream of a given UAS and promoter region (Deitcher, 2023). This paper explores the imaging of five distinct transgenic families of these genetically modified larvae provided by Dr. David Deitcher. 1) The fusion of GFP with a protein found selectively in sensory neurons, such as Nav1.8 (Akopian et al, 1996). 2) The fusion of GFP to a protein involved in the Na^+/K^+ pump in the ventral ganglion, such as ATP-α or nrv2 (Roy et al., 2012). 3) The fusion of a specific green, fluorescent calcium indicator, GCaMP, to proteins required to glutamate transport in motor neuron cells, such as Vglut (Daniels et al., 2004). 4) The fusion of

GCaMP to the protein Vesicular Glutamate Transporter (Vglut) in the ventral ganglion. 5) The fusion of a GCaMP to octopamine, insulin-related proteins surrounding the muscle region, such as DILP2 (Tao et al., 2019).

Fluorescence Microscopy

A constructed fluorescent microscope was utilized for the purpose of this imaging using standard optics lenses, filters, and light sources. Figure 1 demonstrates a labeled diagram of the entire microscope unit and LED source (Johnson, 2025). The optical breadboard connected to the CMOS camera and housing by the aluminum column. Both the linear stage and precision lab jack were utilized for the proper positioning of sample preparations within the optical range. An external LED controlled the light provided to the optical housing. Provided light strikes against the dichroic mirror within the optic house and is deflected toward the sample. Once the blue light reaches the sample, specific regions, known as fluorophores, may experience electronic excitement, emitting photons (Figure 11). Light then emitted from the sample would be able to pass through the dichroic mirror and transmit to the CMOS camera. For green fluorescent protein, we observe that excitation light was determined to be approximately 485 nm and emission light was 515 nm (Figure 2)

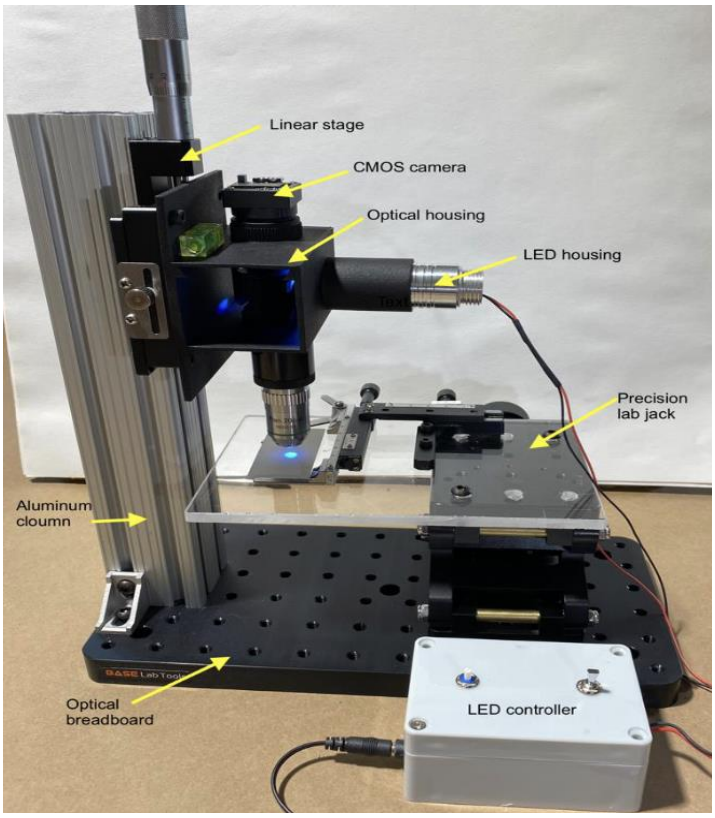


Figure 1. *depicts the complete build of the fluorescence microscope used for all imaging detailed in this experiment. Image derived from Johnson, 2025.*

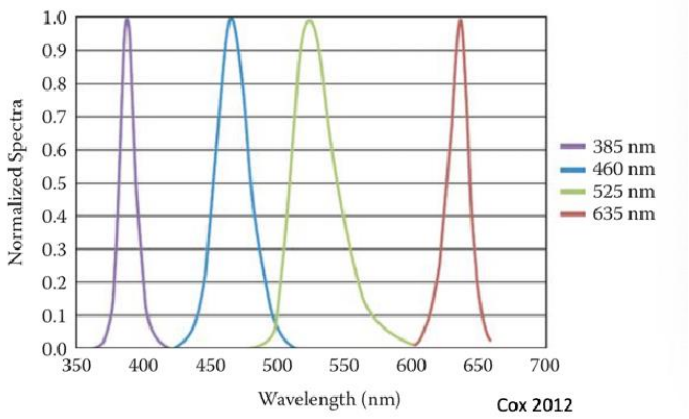


Figure 2. *demonstrates the relative spectral distribution of each distinct color. We observe that the blue light of excitation is observed at approximately 460 nm, the maximum wavelength of green light was 525 nm, and that of red light was 635 nm (Cox, 2012).*

Image Acquisition

The CMOS camera was connected to the computer via USB and operated using Pylon software. The Basler camera method was selected, and the camera settings were initialized to continuous under Analog Control and Acquisition Control. Gain and Exposure Time parameters were adjusted manually to optimize image resolution.

Drosophila fly larva were maintained semi-intact preparation, preserving key nervous system and muscular tissues. Dissections were performed by Dr. Bruce Johnson and the 2025 Teaching Staff of *BIONB 4910: Principles of Neurophysiology*. Each of the five transgenic preparations were obtained in semi-intact form and placed approximately 500 µm from the objective lens.

Preparations 1 and 2 were imaged without any environment manipulation. Preparation 1 was imaged adjacent to the larvae wall and preparation 2 was focused directly on the ventral ganglion. Preparation 3, containing GCaMP-tagged motor neurons, was electrically simulated through an external neuron with a direct synaptic connection to the ventral. The ventral nerve cord and central pattern generator (CPG) of the preparation had been removed, causing fictive locomotion. The fluorescence of a specific ganglion was recorded before stimulation, during stimulation, and after stimulation.

Baseline fluorescence of both transgenic preparations 4 and 5 were obtained in standard saline solution. Then, the standard saline was replaced with saline with an elevated concentration of potassium. The standard saline had a concentration of 5 mM KCl, while the replacement saline had 90 mM KCl (Table 1). The fluorescence was recorded immediately after replacement and then again after several minutes. Preparation 4 imaged the ventral ganglion while preparation 5 focused on the muscle directly.

All images captured from Pylon were processed using standard ImageJ software. The background of images was

removed by subtracting a mean pixel intensity of the background. Then, a Green Lookup Table (LUT) effect was implemented to correspond to the applied GFPs. A 100 µm Scale Bar was added to each image in the bottom right corner.

Component	Normal Saline (mM)	High K^+ Saline (mM)
NaCl	130	45
KCl	5	90
Sucrose	36	36
$MgCl_2$	2	2
$CaCl_2$	2	2
HEPES (pH 7.3)	5	5

Table 1. *details the respective concentration of various components in both the standard and high K^+ saline solutions. This becomes highly important for the manipulation of preparations 4 and 5.*

RESULTS

The initial preparation (1) features the fusion of protein present selectively found in sensory neurons to GFP (Figure 3). The larval wall was imaged via fluorescent microscopy. We observe that the larger, round fluorescent regions correspond to a cluster of cell bodies. Their morphology, specifically the round shape, suggests that these are sensory neurons. Additionally, there is a robust and extensive neuronal network of axons and dendrites exiting the cell bodies.

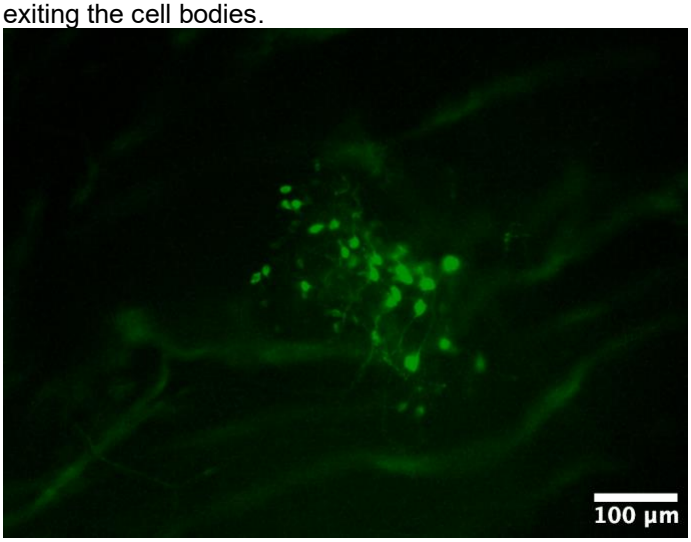


Figure 3. *details the GFP tagged proteins in sensory neurons (green) found in the larval wall. When exposed on the external site, firing of neurons exhibit.*

A similar preparation (2) was applied for proteins found in the Na^+/K^+ pump in the ventral ganglion (Figure 4). In Panel A, we observe the fluorescing of these proteins within the cell bodies through a measurement with low exposure time. By increasing the exposure time, we were able to observe the fluorescing of this tagged protein in the axons leaving the cell bodies of the ventral ganglion (Panel B). We observe an overall loss of resolution but gain the ability to image these diverging axons.

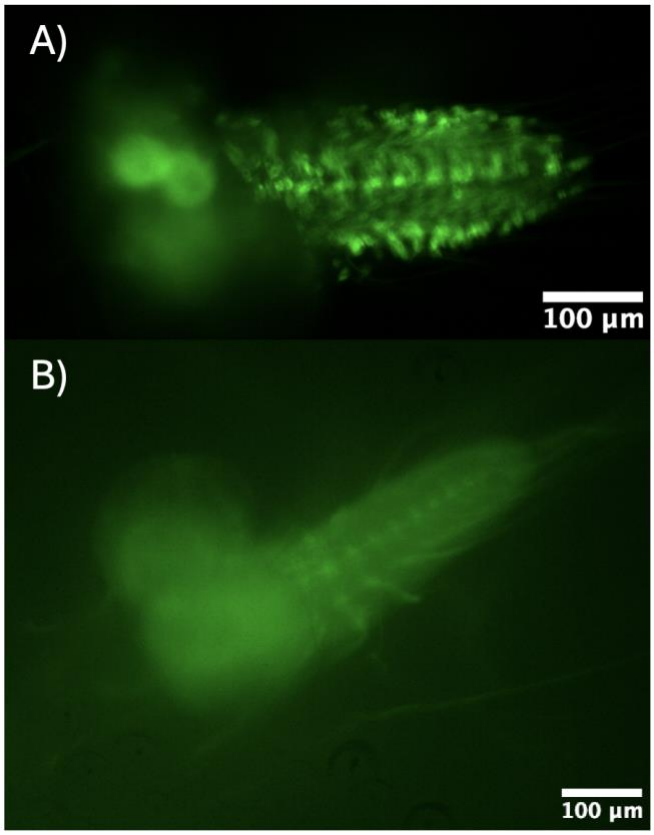


Figure 4. *demonstrates the GFP tagged Na^+/K^+ pump. We observe the presence of this protein specific to this pump fluorescing throughout the ventral ganglia in panel A. The presence of this protein is also found in axons leaving this ganglion in panel B, demonstrating the mechanism of the pump in Action Potential formation.*

The third preparation (3) relied on the fusion of GCaMP7b to Vglut in the ventral ganglion. The mean raw fluorescence for the central, prominent ganglion was quantified via histogram distribution (red arrow, Figure 5). We observed an increase in the GCaMP fluorescence following to electrical stimulation. Before stimulation (Panel A, Figure 5), there was an observed a mean raw fluorescence of 13.389 RFU. After stimulating the external neuron and enacting fictive locomotion (Panel B, Figure 5), the mean raw fluorescence increased to 15.064 RFU.

We were able to plot the raw fluorescence for the entire image range for the baseline, during the active fictive locomotion, and after the fictive locomotion finished (Figure 6). We observe that the action of fictive locomotion corresponded to an increase of raw fluorescence from 153.481 RFU to 172. 541 RFU. The expression of the protein ultimately returned to baseline as the raw fluorescence decreased below the original value.

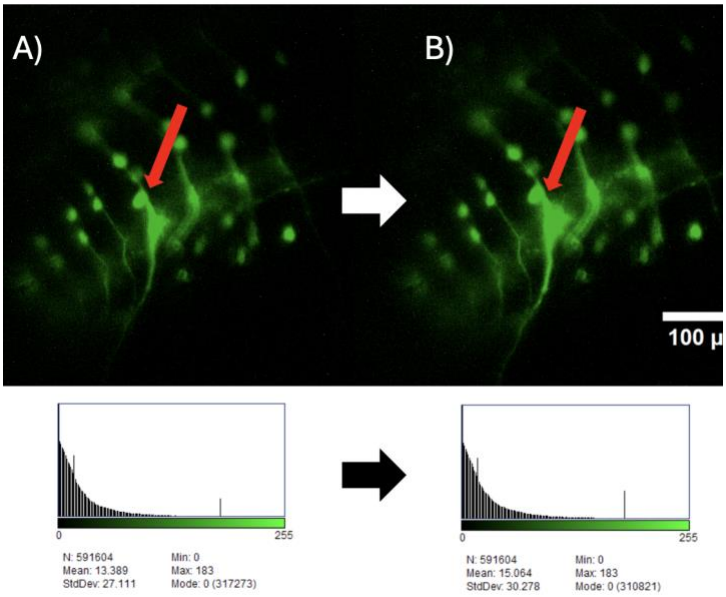


Figure 5. demonstrates the behavior of fictive locomotion in the ventral ganglia. Panel A and B contain both a GCaMP-fused protein in the ventral ganglia and the respective intensity histogram. Panel A is the baseline obtained before any stimulation was applied. Panel B details the response of the ventral ganglia in response to external neural stimulation.

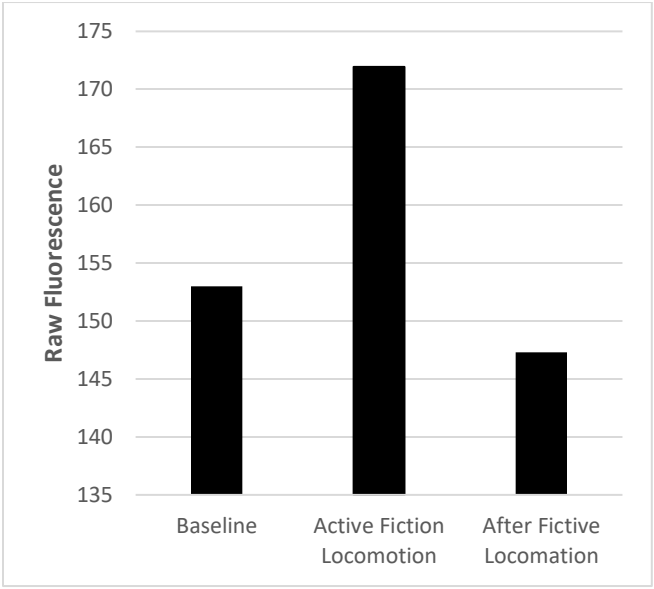


Figure 6. *expresses the quantification of total Raw Fluorescence at three distinct time points. We observe a raw fluorescence of 172.541 RFU during fictive locomotion and a raw fluorescence of 153.481 RFU in the baseline.*

The same GCaMP-tagged procedure was provided by Dr. David Deitcher for Vglut in the ventral ganglion in preparation 4. In Figure 7, we observe the increase in fluorescence throughout the entire cell as a function of increasing extracellular K^+ concentration. The mean raw fluorescence was determined as the average fluorescence value of a region directly corresponding to the ventral ganglion. In panel A, we observe basal expression of these glutamate transport proteins, quantified by a mean raw fluorescence of 44.852 RFU. Immediately after the introduction of an extracellular solution of high K^+ concentration, we observe an observable increase in fluorescence, quantified by an increased mean raw fluorescence of 73.974 RFU.

The entire image was analyzed in response to raw fluorescence and relative fluorescence (Figure 8). Raw fluorescence quantifies the relative saturation of a given pixels. Relative fluorescence compares the raw fluorescence at each image to that of the baseline. We observe a sharp increase following the introduction of high potassium saline, followed by a return to the baseline fluorescence after 540 seconds.

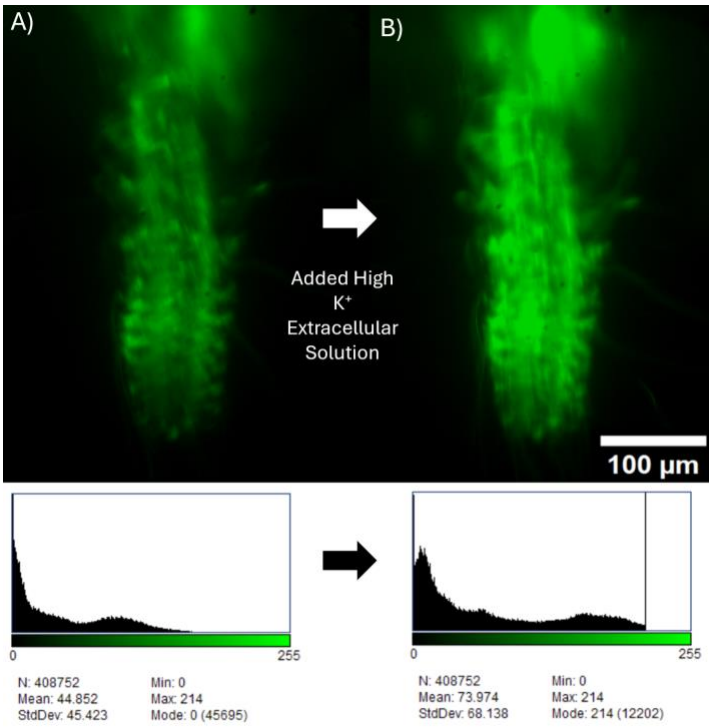


Figure 7. *demonstrates the baseline of fluorescence activity before K^+ was added to prep in Panel A. This has a mean raw fluorescence of 44.852 RFU. Panel B demonstrates fluorescence directly after high K^+ saline was added, with a mean raw fluorescence of 73.974 RFU.*

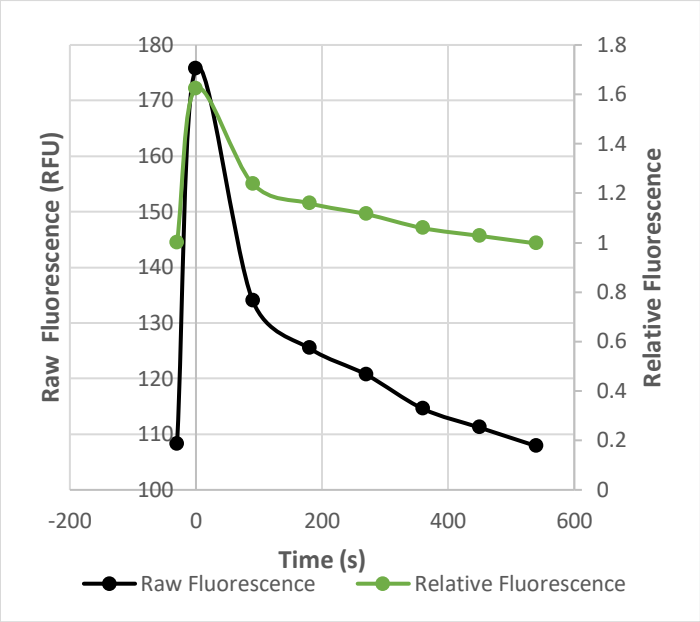


Figure 8. *demonstrates both the raw fluorescence and the relative fluorescence in the ventral ganglia as a function of time during the introduction of high K⁺ saline. We observe that there is a rapid increase in raw fluorescence, followed by a gradual decrease back to the observed baseline.*

In the final preparation (5), a GCaMP fused octopamine-transport protein the neurons (DIPL2) to form the complex TDC>DILP2-GFP. The muscle sites were directly imaged. The imaging highlights the presence of distinct circular collections of fluorophores in the form of boutons and axons. Figure 9, Panel A represents a baseline of fluorescence with distinctive octopamine transport. This baseline consists of a mean raw fluorescence of 11.337 RFU. Two minutes after applying this high concentration of potassium in saline, we observe significantly less fluorescence behavior in Panel B. This decrease in fluorescence is quantified by the mean raw fluorescence of 5.412 RFU and observably less neurotransmitter release. There appears to be several boutons and axons that are not imaged in Panel B that are significantly fluorescing in baseline recording.

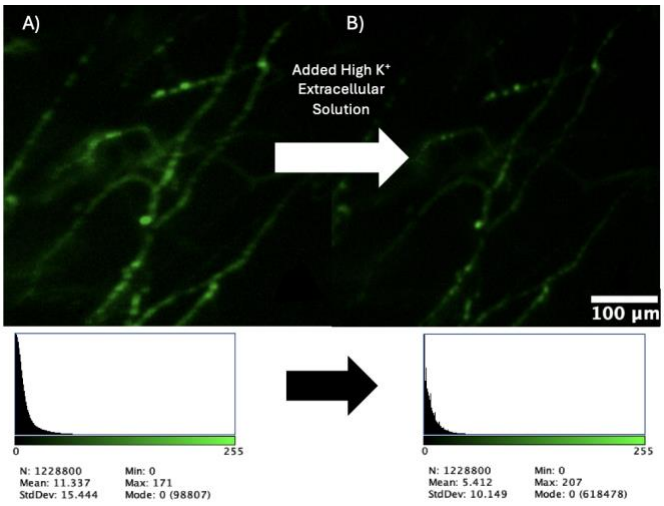


Figure 9. *demonstrates the action of the addition of High K⁺ saline to imaging directly in the muscles. Panel A details the baseline and has a mean raw fluorescence of 11.337 RFU while Panel B quantifies the moment directly after introduction as 5.412 RFU.*

DISCUSSION

Fluorescence microscopy utilizes fluorophores, such as GFP-bound proteins, to image biological systems. The microscope build demonstrates the specifics of a fluorescent microscopy, using a dichroic mirror and filters to properly categorize excitation and emission light (Figure 1). Figure 10 demonstrates the respective pathway of both excitation and emission light. In this experiment, we utilize GFP which is excited at 485 nm and emitted at 515 nm (Johnson, 2025). These respective wavelengths correspond to the excitation of GFP via blue-VIS light and emission of green-VIS light (Figure 2). Blue light is produced by the LED light-source, deflected off the dichroic mirror and directed to the sample (Figure 10, Panel A). Panel B demonstrates the optical behavior of a dielectric interference filter, where light is selectively transmitted or deflected based on constructive interference across layers of differing refractive indices (Stuurman, 2025). The emitted light expelled by the sample (green) can transmit pass the dichroic mirror as it selectively experiences constructive interference.

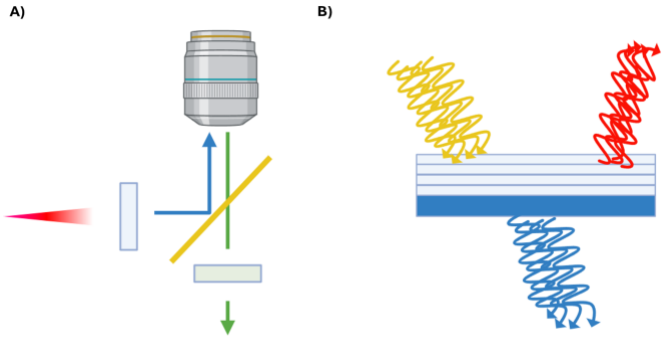


Figure 10. *demonstrates the optimal interactions between light sources and waves in the fluorescence microscope developed. In panel A, the light source generates blue-VIS light after passing an excitation filter. We also contain a dichroic mirror (yellow) and green light (green). The excitation filter is expanded in Panel B.*

Jablonski diagrams demonstrate the behavior of electrons on a molecular level to drive fluorescence (Figure 11). Information regarding these mechanics were derived from Stuurman, 2019. When higher-energy blue light illuminates the sample, it excites an electron from the ground state (S_0) to the first electronic excited state (S_1) within femtoseconds (blue, Figure 11). Following excitation, the electron undergoes internal conversion, relaxing to the lowest vibrational level of S_1 (v' = 0) within picoseconds (red). Finally, the electron returns to the ground state by emitting a photon of lower energy, green light on the nanosecond timescale (green). The green light is then able to pass through the dichroic filter as it has a fundamentally longer wavelength than the blue excitation light. We

observe this fluorescent behavior in all neurons that contain GFP-bound proteins. The CMOS camera can detect this green light emitting from the sample and generate a respective image.

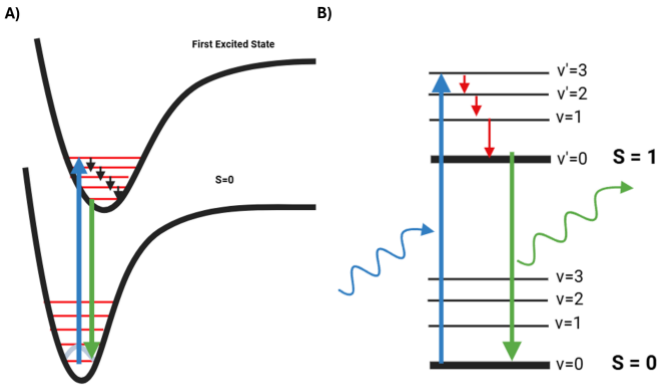


Figure 11. *details the electronic and vibrational transitions between the ground state and first electronic state of the GFP-bound proteins. Blue arrows are used to indicate like in the blue-VIS spectrum, consistent for red and green arrows.*

Fluorescence Microscopy was applied to various living preparations to quantify and image neural activity. We first observe two preparations of standard GFP-fused proteins in various, specified regions of the Drosophila. The GFP used in these preparations were found in a constantly activated form. This means that upon cellular expression of these various proteins, the protein would always fluoresce when exposed to blue light.

The first preparation (1) consisted of the imaging of a region of sensory neurons on the larval wall. In Figure 3, we observe distinct clusters of fluorophores reflecting various cell bodies. Their round shape and distinct neural network suggest that these sensory neurons are experiences strong activation. *In Nociceptive neurons protect Drosophila larvae from parasitoid wasps*, Hwang et al. explores the mechanism of class IV multidendritic (md) neurons as nociceptors. In response to significant thermal or mechanic pressures, these class IV md neurons essentially indicate the feeling of pain, eliciting a respective defense mechanism. There sensory neurons are found specifically on the larval wall to detect for external pressures, such as the presence of predation. Drosophilae contain multiple mechanosensitive and thermally sensitive channels to observe of danger – this paper addresses the presence of a parasitoid wasp. In response to this danger, these sensory neurons experience accelerated firing, eliciting a corkscrew, rolling mechanism in defense. The GFP stained images in Figure 3 provides specific images to specifically target these class IV md neurons. We observe that the presence of these nociceptive neurons is evolutionarily favorable to avoid parasitic wasp attacks.

A similar GFP protein was tagged to a protein highly conserved in the Na^+/K^+ pump in the ventral ganglion. In Figure 4, we observe the presence and action of this pump in two distinct images. We observe the function of this pump in soma to maintain the electrochemical gradient required for a polarized membrane (Panel A, Figure 4). These cell bodies exhibit strong fluorescence in the living prep, corresponding to the pumping of ions to establish resting potential or the initiation of action potential formation via a Na^+ gradient (Clausen, 2005). Similarly, we observe the effects of action potential propagation in Panel B. As an action potential traverses from the soma to the synaptic cleft, the membrane requires both depolarization and hyperpolarization. The specificity of fluorescence microscopy imaging enables the determination that the sodium-potassium pump is critical for both the initiation and propagation of action potentials.

Another GFP variant, GCaMP was utilized to explore changes to the synaptic behavior of neurons through its strong association with Ca^{2+}. Similarly to a traditional GFP, the gene encoding for GCaMP is selectively inserted adjacent to a specific protein of interest. However, we observe that within this state of Protein-GCaMP, the molecule does not fluoresce, and electrons cannot be readily excited. Rather, the Protein-GCaMP must bind with Ca^{2+}, eliciting a conformational change in tertiary protein structure (Johnson, 2025). In Figure 12, Panel A demonstrates an unbounded complex that does not fluoresce. Conversely, Panel B, elicits a conformational change upon the binding of Ca^{2+}, that allows electrons to be easily excited. The formation of action potentials regularly elicits the influx of calcium at the pre-synaptic site. The incoming calcium drives the formation and exocytosis of neurotransmitter vesicles into the post-synaptic site. In binding GCaMP to ai protein involved in this neurotransmitter release, we were able to directly target the specific cell type and region of interest. Therefore, when neurotransmitter release is prominent and action potentials are propagating, a higher expressing of active Protein-GCaMP-Ca^{2+} is fluorescing.

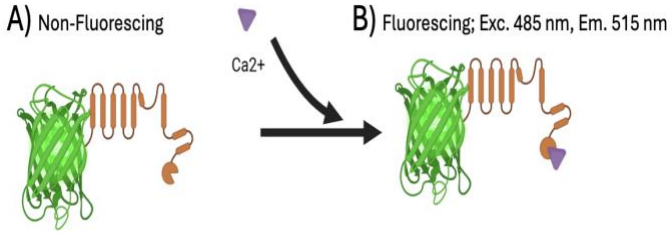


Figure 12. *details the conformational change associated with the binding of Ca^{2+} to the Protein-GCaMP complex. This binding results in the fluorescence of the unit at an excitation wavelength of 485 nm and an emission of light at 515 nm.*

In the third preparation (3), glutamate-transport proteins were tagged with GCaMP7b, forming Vglut-GCaMP7b. After stimulating the external neuron, we observe a substantial increase in raw fluorescence and the mean raw fluorescence of the central ganglion (Figure 5,6). An increase in fluorescence of the Vglut-GCaMP7b fusion protein suggests an increase in the concentration of

Calcium. With increased levels of Ca^{2+}, the proteins would be more predominantly found in the Vglut-GCaMP7b-Ca^{2+} bounded state which fluoresces after being excited by the provided blue light.

The neuron that experiences external stimulation generates an action potential that propagates through an axon to a pre-synaptic site. At this pre-synaptic site, the depolarization of the membrane increases the permeability of Ca^{2+} to the membrane. Increased permeability and requirement for the vesicle formation mechanism results in increasing amounts of Calcium in the pre-synaptic terminal. Depolarization of the membrane allows for the opening of voltage-gated channels just as P/Q-type and N-type currents (Wheeler et al., 1994). Calcium is thus required for the formation of vesicles required to transport neurotransmitters, such as glutamate, out of the cell. The external stimulation traditionally corresponds to a standard motion action, suggesting the propagation of excitatory neurotransmitters such as glutamate. However, in this preparation (3), the central nerve cord and CPG have been removed, meaning that no motion is demonstrated. The result is fictive locomotion such that the larvae ventral ganglion will continue to promote these action potentials to motor neurons, but the CPG connection is missing completely. We observe that the firing to initiate motion continues, driving Calcium into the presynaptic site and resulting in neurotransmitter release. As the substrate concentration increases, standard receptor-ligand kinetics suggest that more Vglut-GCaMP7b-Ca^{2+} complexes are found. In Figure 5, we observe a significantly greater mean raw fluorescence in the central ganglion during fictive locomotion (15.064 RFU) than during the baseline recording (13.389 RFU). This increase in fluorescence was a direct result of AP-driven calcium influx resulting in more fluorescence-bound states. Ultimately, after stimulation is released, the Calcium will diffuse out the pre-synaptic site, and baseline fluorescence is reestablished.

The fourth preparation (4) resulted in the formation of fusion protein Vglut-GCaMP to image a protein strongly associated in glutamate transport in the ventral ganglia. We initially provided a baseline recording of fluorescence in the ventral ganglia with standard saline solution. We obtained a mean raw fluorescence of 44.482 RFU throughout the ventral ganglion (Figure 7). This suggests that there is a basal rate of action potential formation in the ventral ganglion when exposed to standard saline. Action potentials propagate, resulting in the opening of voltage gated currents such as P/Q-type and N-type, increasing the pre-synaptic calcium concentration (Wheeler et al., 1994). The fluorescence of these Vglut-GCaMP-proteins is completely dependent on their binding properties to Ca^{2+} (Figure 12).

When the respective potassium chloride concentration in saline solution was increased from 5 mM to 90 mM, we observed a distinctive increase in fluorescence. The ventral ganglion imaging an increase in mean raw fluorescence to 73.974 RFU, suggesting a higher concentration of calcium in the pre-synaptic site. As the extracellular concentration of potassium increased, the driving force for potassium shifted. Generally, the intracellular concentration of K^+ of approximately 140 mM, suggesting a driving force of K^+ out of the cell (Singh et al., 1989). In Equation 1, the driving force is quantified both subject to standard and high K^+ concentrations. Under standard 5 mM KCl saline solutions, we observe a driving force of 13.39 mV corresponding to the outflux of K^+. This driving force worked to hyperpolarize the membrane and combats against the passing threshold membrane potential. However, as the extracellular concentration increases from 5 mM to 90 mM, the driving force reverses signs to -55.60 mV. With this increased potassium concentration, we now see an influx of potassium, working to depolarize the membrane. This influx of positive charge drives membrane potential toward the AP threshold, proving the neuron far more excitable.

A)

$$V_{K^+} = \frac{RT}{F}\ln\frac{[K^+]_{out}}{[K^+]_{in}} = \frac{RT}{F}\ln\frac{5\ mM}{140\ mM} = -78.38\ mV$$
$$V_{DF} = V_m - V_{K^+} = -65\ mV + 31.95\ mV = 13.39\ mV$$

B)

$$V_{K^+} = \frac{RT}{F}\ln\frac{[K^+]_{out}}{[K^+]_{in}} = \frac{RT}{F}\ln\frac{90\ mM}{140\ mM} = -10.40\ mV$$
$$V_{DF} = V_m - V_{K^+} = -65\ mV + 10\ mV = -55.60\ mV$$

Equation 1. *details the determination of the reversal potential and driving force for both 36 mM extracellular saline concentration (A) and 90 mM extracellular saline concentration (B).*

The High K^+ conditions therefore drive a significantly greater frequency of AP firing. Often, the neurotransmitter release cannot maintain this rate of firing, resulting in a buildup of residual Ca^{2+} in the pre-synaptic terminal (Zucker et al., 2002). Through an increases number of collisions between Ca^{2+} and Vglut-GCaMP, there are an increased proportion of fluorescing proteins in the pre-synaptic site. As a result, we observe this increased fluorescence within the ventral ganglion. In Figure 8, we observe an immediate increase in raw fluorescence. Over the course of 9 minutes, we observe the raw fluorescence decay to the initial, baseline fluorescence. Due to intense, consistent action, the VG-Ca^{2+} channels inactivate over time and the residual concentration of Ca^{2+}approach the previous, basal value. This results in a delayed return to the original fluorescence observed as there are fewer Vglut-GCaMP-Ca^{2+}complexes.

The final preparation (5) was prepared by fusing a GCaMP protein to an octopamine-transport protein Tdc2 DLIP2-GFP. DILP2 is a peptide structurally similar to human insulin (Brown et al., 2020). We imaged both during baseline and two minutes after high K^+ was administered. Specifically associated to the vesicle formation region of the neuron and being adjacent to high concentration of calcium, the protein fluoresces in regions experiencing presynaptic release of neurotransmitters. In Panel A, we observe significant release transport of octopamine both in

boutons and along the axons. This suggests the presence of an advanced neural network directly along the larvae muscle tissue. The neural network was imaged and analyzed using ImageJ, indicating an average raw fluorescence of 11.337 RFU for the larvae in 36 mM KCl.

Following the introduction of 90 mM KCl saline, there was an observed decrease in fluorescence. ImageJ processes indicated a mean raw fluorescence of 5.412 RFU, suggesting a significantly lowered proportion of Ca^{2+} bound state proteins. Supported by Kuromi et al., 1998, these results suggest the mechanism of K^+ facilitating vesicle release through cycling and exocytosis. As K^+ accelerates the formation of vesicles which release neurotransmitters, these dense core vesicles (DCVs) are identified as a pivotal component of the observed decrease in fluorescence. When DILP2 is expressed under Tdc2-GAL4, we observe that octopaminergic neurons will package these DILP-GFP proteins into DCVs (Tao et al., 2019). Many neuropeptides, such as the octopamine-associated proteins are present in the release of the neurotransmitters. Thus, as the vesicle release increases, the respective concentration of neuropeptides, such as the one bonded to DLIP2-GFP decreased. Rather, these proteins are dispersed into the extracellular solution and the overall fluorescence decreases.

We observe that imaging techniques, specifically fluorescent microscopy has provided a representation into a population of cells. Throughout the imaging of drosophila fly larvae, the imaging largely concentrated on a specific population of cells. Imaging results often represent the contribution of thousands of neurons, contributing to poor resolution and an overall blurring effect from several adjacent fluorophores. Similarly, fluorescence microscopy often results in photobleaching such that continuous excitation ultimately degrades the fluorophores fluorescence abilities (Snapp, 2009). However, this imaging enables the genetic modification of drosophila such that the specificity of recording can be highly accurate. This manipulation of recording, enabling for the selective imaging of various proteins and thus neural activity. This high specificity of recording is extremely difficult to accomplish in electrophysiological recordings.

Electrophysiological recording specificity requires the identification of which cell or region electrode-based recordings are performed. Comparison on the various applications and implications of recording techniques were derived from Luo, 2020: *Principles of Neurobiology. As* the tip of a standard intracellular electrode is too small to image in a standard light microscope, it is highly unreasonable to assume high specificity of recording. However, these recordings produce a degree of sensitivity and temporal resolution that optical imaging cannot obtain. Electrode-based recording enables the recording of neurons or synaptic regions at any depth or location within a sample. These electrophysiological recordings generally fixate on a single neuron or synapse, without being blurred by the electrical noise of adjacent neurons. This allows for extremely sensitive changes in membrane potential to be recorded on that time scale of milliseconds. While fluorescent microscopy is adequate for imaging large-scale patterns, the propagation of action potentials requires the sensitivity and resolution of electrode-based measurement.

While fluorescent microscopy is necessary for these highly specific, selective recording of neural activity, there is a distinctive loss of depth of recording. Fluorescence requires the penetration of blue light to the fluorophores of interest. When we excite with light blue light, we observe that there is distinct absorbance of the tissues. In Figure 13, we observe that both Melanin and Haemoglobin have substantial absorption in the blue-VIS spectrum. Therefore, the excited light cannot reach deep tissues and there is dampened fluorescent effects. Another mechanism of light dispersing before reaching the tissue of interest is light scattering. We observe that an incident photon will instantaneously be absorbed by a given molecule and emitted with the same energy in any random direction. Particularly for blue light, this scattering of light with atmospheric gases such as CO_2 and N_2 becomes a key loss of light within 1-Photon Fluorescent Microscopy.

An observed solution to this limited depth of fluorescent microscopy is largely addressed using 2-Photon Fluorescent Microscopy (Richardson, 2019). To excite an electron in the GFP-bound molecule, an energy corresponding to blue light is required. Rather than supplying this energy with one short-wave photon, this approach is categorized by its delivery of two longer, red-light photons (Figure 14). As depicted in Figure 13, we observe a sharp decrease in tissue absorption of light through pigments of Melanin and Haemoglobin comparing the blue and red sections of the visible spectrum. Utilizing two red-light photons delivered simultaneously, the electron is excited from the electronic ground state to the equivalent excited state (Panel B, Figure 14). Through a lower absorption coefficient and a weaker tendency of scattering due to atmospheric gases, the longer wavelength light used two photon microscopies can drive deeper penetration into the sample. However, we observe that both 1-photon and 2-photon fluorescence emit green light of the same frequency (Figure 14). Limiting the effects of 2-Photon Fluorescence Microscopy, we observe that this emitted light will be absorbed by the tissue as a substantial rate as it leaves the sample (Figure 13).

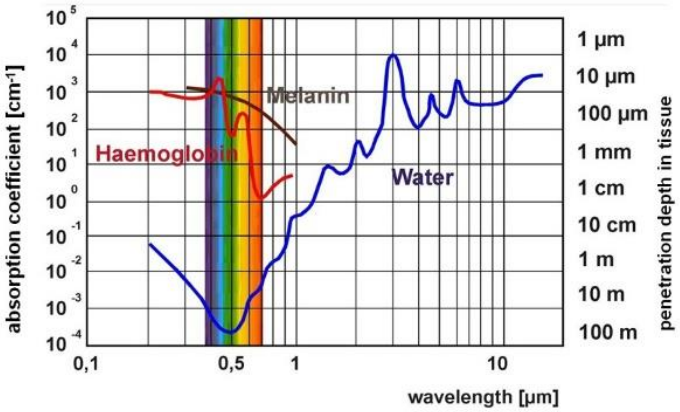


Figure 13. *demonstrates the relative absorption coefficient of Melanin (brown), Haemoglobin (red) and water (blue) as a function of their wavelength. We observe that Haemoglobin and Melanin decrease as a function of wavelength. Image derived from Schoelle et al., 2019.*

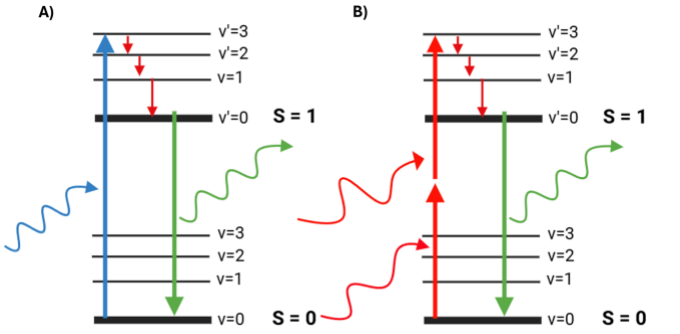


Figure 14. *demonstrates the Jablonski diagram for both 1-Photon Fluorescence (Panel A) and 2-Photon Fluorescence (Panel B).*

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